Advanced AI Campus Hackathon

Smart Campus, Better Society

Expanded Problem Statement Catalogue — Revised

Difficulty scale: 1/10 = introductory, 10/10 = extremely challenging.

Evaluation suggestion: Social impact 25% | Technical implementation 25% | Innovation 20% | UI/UX 15% | Practical feasibility 15%

Original Problem Statements

PS-01 Student Dropout Early Warning System — 8/10

Use simulated attendance, academic performance, and engagement data to identify students who may need additional support. Create explainable, privacypreserving predictions and equitable intervention recommendations.

PS-02 Intelligent Campus Disaster Response — 9/10

Build a platform to simulate fires, floods, earthquakes, and other campus emergencies. Implement dynamic evacuation routing that adapts to hazards, crowd density, and accessibility needs.

PS-03 Campus Food Demand Prediction — 6/10

Forecast daily mess and canteen demand using historical meals, holidays, exam schedules, and attendance estimates. Reduce food waste while avoiding shortages under uncertain demand.

PS-04 AI Campus Accessibility Auditor — 8/10

Analyze campus maps and infrastructure images to identify obstacles faced by people with disabilities. Recommend accessible routes and prioritize infrastructure improvements.

PS-05 Campus Cybersecurity Threat Detection — 9/10

Create a dashboard that monitors a simulated campus network for suspicious activity. Detect anomalous logins, phishing indicators, and attack patterns with low false-positive rates.

PS-06 Campus Resource Allocation Optimizer — 9/10

Allocate classrooms, laboratories, equipment, and study areas based on competing requirements. Solve scheduling conflicts under capacity, fairness, accessibility, and availability constraints.

PS-07 Privacy-Preserving Harassment Pattern Detection — 9/10

Design a confidential reporting system that helps authorized staff identify potentially related incidents. Find patterns without exposing complainants, making automated accusations, or bypassing human review.

PS-08 Campus Energy Optimization — 8/10

Predict electricity demand in hostels, labs, and academic buildings. Use occupancy, weather, and simulated meter data to reduce consumption while maintaining essential services.

PS-09 Campus Misinformation Detection — 8/10

Help students check potentially false college announcements against verified institutional information. Provide evidence-linked assessments, uncertainty labels, and source traceability.

PS-10 Emergency Medical Coordination — 9/10

Coordinate verified emergency requests with campus responders and healthcare contacts. Optimize assignment and routing using incident severity, availability, and travel time.

PS-11 Campus Sustainability Decision Engine — 9/10

Evaluate the environmental impact of campus operations across energy, water, waste, and transport. Recommend a cost-constrained portfolio of improvements using multi-objective optimization.

Additional High-Impact Problem Statements

PS-12 AI-Based Mental Health Support and Crisis Detection — 9/10

Develop a privacy-preserving platform that identifies students at risk of severe stress, burnout, or self-harm using voluntary surveys, academic patterns, and help-seeking behavior. Detect risk without diagnosis, protect privacy, minimize false alarms, and connect students to verified support.

PS-13 Women's Safety and Safe Route Optimizer — 8/10

Recommend safer walking routes using lighting, crowd density, emergency phones, security coverage, and historical incident data. Optimize routes in real time while protecting sensitive location data and avoiding victim-blaming.

PS-14 AI Sign-Language and Speech Accessibility Assistant — 9/10

Create a real-time tool that converts sign language to text or speech and spoken language into visual instructions. Support diverse signing styles, regional languages, limited training data, and low-latency operation.

PS-15 Assistive Navigation for Visually Impaired Students — 9/10

Develop an AI navigation assistant using computer vision, maps, voice instructions, and obstacle detection. Provide safe, accurate, real-time guidance and identify temporary obstacles.

PS-16 Scholarship and Financial-Aid Eligibility Engine — 8/10

Identify students who may qualify for scholarships, fee support, food assistance, or emergency funds. Match complex eligibility rules while protecting financial data and preventing discrimination.

PS-17 Campus Food Rescue and Redistribution Platform — 8/10

Predict surplus food from hostels and canteens and coordinate safe redistribution to students, shelters, or community organizations. Forecast surplus, optimize collection routes, and monitor food safety.

PS-18 Student Accessibility Accommodation System — 8/10

Convert confidential disability-related accommodation requests into coordinated academic, transportation, examination, and housing support. Prioritize urgent needs, detect service gaps, and provide explainable recommendations.

PS-19 Human Trafficking and Exploitation Risk Awareness System —

9/10 Detect suspicious patterns in simulated campus or community reports without automatically accusing individuals. Analyze weak signals, preserve anonymity, reduce bias, and require human verification before escalation.

PS-20 AI-Based Academic Integrity and Fairness Monitor — 9/10

Detect suspicious academic behavior while protecting students from unfair automated punishment. Distinguish collaboration from cheating, detect bias, provide evidence, preserve privacy, and keep humans in the decision loop.

PS-21 Community Blood and Organ Donation Coordinator — 9/10

Predict blood demand, verify donors, match compatible donations, and coordinate urgent requests. Handle real-time matching, medical privacy, availability uncertainty, and emergency prioritization.

PS-22 Campus-to-Employment Opportunity Matcher — 8/10

Connect students—especially first-generation, rural, or economically disadvantaged learners—with suitable internships, jobs, and skill-development programs. Match skills to opportunities, identify skill gaps, explain recommendations, and prevent bias.

Recommended Highest-Impact Choices

For maximum technical challenge and social value, consider PS-12, PS-14, PS-15, PS-19, PS-21 and PS-22 These combine AI complexity, measurable social benefit, safety considerations, privacy requirements, and strong opportunities for innovation.

General Organizer Notes

• Provide common or simulated datasets and surprise test cases.

• Judge working functionality, safety, fairness, and measurable outcomes—not chatbot integration alone.

• Require human review for high-risk decisions involving health, safety, harassment, finances, or academic consequences.

• Teams should document limitations, false positives, privacy protections, and responsible-AI safeguards.